

Guneyasu Ayse

TD 1 :

Part 1/2/3/4 : It's been done.

Part 5 :

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ pwd
/workspaces/esilv-marketpulse-g01-t07
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ ls
CONTRIBUTING.md  TEAM TEMPLATE.md  data  labs  requirements.txt  td1.pdf
README.md        config  evidence  readiness  src
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ ls -la
total 512
drwxrwxrwx+ 9 codespace root    4096 Oct  7 23:08 .
drwxr-xrwx+ 5 codespace root    4096 Oct  7 23:08 ..
drwxrwxrwx+ 8 codespace root    4096 Oct  7 23:09 .git
-rw-rw-rw-  1 codespace root     178 Oct  7 23:08 .gitignore
-rw-rw-rw-  1 codespace root    2943 Oct  7 23:08 CONTRIBUTING.md
-rw-rw-rw-  1 codespace root   10270 Oct  7 23:08 README.md
-rw-rw-rw-  1 codespace root    1446 Oct  7 23:08 TEAM TEMPLATE.md
drwxrwxrwx+ 2 codespace root    4096 Oct  7 23:08 config
drwxrwxrwx+ 3 codespace root    4096 Oct  7 23:08 data
drwxrwxrwx+ 2 codespace root    4096 Oct  7 23:08 evidence
drwxrwxrwx+ 2 codespace root    4096 Oct  7 23:08 labs
drwxrwxrwx+ 2 codespace root    4096 Oct  7 23:08 readiness
-rw-rw-rw-  1 codespace root     133 Oct  7 23:08 requirements.txt
drwxrwxrwx+ 2 codespace root    4096 Oct  7 23:08 src
-rw-rw-rw-  1 codespace root  455077 Oct  7 23:08 td1.pdf

@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ cd data
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07/data (ayseguneyasu) $ pwd
/workspaces/esilv-marketpulse-g01-t07/data
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07/data (ayseguneyasu) $ ls
sample
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07/data (ayseguneyasu) $ cd sample
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07/data/sample (ayseguneyasu) $ pwd
/workspaces/esilv-marketpulse-g01-t07/data/sample
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07/data/sample (ayseguneyasu) $ ls
bloomberg_reference_expected.json  bloomberg_reference_sample.json  instruments.json  prices.csv
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07/data/sample (ayseguneyasu) $ cd ../../
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ pwd
/workspaces/esilv-marketpulse-g01-t07
```

Part 6 :

```
@ayse3993 →/workspaces/esilv-markets-pulse-g01-t07 (ayseguneyasu) $ cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
```

The instrument is Apple Inc. (AAPL), traded on the NASDAQ.

The benchmark is the S&P 500 (SP500).

Their names and tickers are stored in data/sample/instruments.json, under the instrument and benchmark sections.

```
@ayse3993 →/workspaces/esilv-markets-pulse-g01-t07 (ayseguneyasu) $ head data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
```

```
@ayse3993 →/workspaces/esilv-markets-pulse-g01-t07 (ayseguneyasu) $ grep AAPL data/sample/prices.csv
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
2026-09-09,AAPL,254.00,255.70,252.80,254.20,38000000
2026-09-10,AAPL,254.30,256.60,253.10,255.10,39500000
2026-09-11,AAPL,254.20,256.20,253.00,254.70,41000000
2026-09-14,AAPL,256.60,258.30,255.40,256.80,42500000
2026-09-15,AAPL,257.20,259.50,256.00,258.00,44000000
2026-09-16,AAPL,256.70,258.70,255.50,257.20,38000000
2026-09-17,AAPL,259.20,260.90,258.00,259.40,39500000
2026-09-18,AAPL,259.30,261.60,258.10,260.10,41000000
2026-09-21,AAPL,259.10,261.10,257.90,259.60,42500000
2026-09-22,AAPL,261.10,262.80,259.90,261.30,44000000
2026-09-23,AAPL,261.20,263.50,260.00,262.00,38000000
2026-09-24,AAPL,263.00,265.00,261.80,263.50,39500000
2026-09-25,AAPL,262.60,264.30,261.40,262.80,41000000
2026-09-28,AAPL,263.30,265.60,262.10,264.10,42500000
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
```

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ grep SP500 data/sample/prices.csv
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,SP500,6623.00,6643.00,6611.00,6628.00,0
2026-09-09,SP500,6640.00,6657.00,6628.00,6642.00,0
2026-09-10,SP500,6642.00,6665.00,6630.00,6650.00,0
2026-09-11,SP500,6642.00,6662.00,6630.00,6647.00,0
2026-09-14,SP500,6663.00,6680.00,6651.00,6665.00,0
2026-09-15,SP500,6664.00,6687.00,6652.00,6672.00,0
2026-09-16,SP500,6663.00,6683.00,6651.00,6668.00,0
2026-09-17,SP500,6683.00,6700.00,6671.00,6685.00,0
2026-09-18,SP500,6684.00,6707.00,6672.00,6692.00,0
2026-09-21,SP500,6683.00,6703.00,6671.00,6688.00,0
2026-09-22,SP500,6703.00,6720.00,6691.00,6705.00,0
2026-09-23,SP500,6704.00,6727.00,6692.00,6712.00,0
2026-09-24,SP500,6715.00,6735.00,6703.00,6720.00,0
2026-09-25,SP500,6713.00,6730.00,6701.00,6715.00,0
2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
```

Part 7 :

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ python --version
Python 3.14.2
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ git --version
git version 2.55.0
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
```

Part 8 : It's been done.

Optional 1 :

Having two aligned series is useful because both the stock and the benchmark have data for the same dates. This makes it easier to compare their performance over time.

```

• @ayse3993 → /workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ grep -c AAPL data/sample/prices.csv
21
• @ayse3993 → /workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ grep -c SP500 data/sample/prices.csv
21

```

Optional 2 :

ls -lh shows the files and folders in the current directory. head -n 5 shows the first 5 lines of the CSV, while tail -n 5 shows the last 5 lines. So, the main difference is that head looks at the beginning of the file and tail looks at the end.

```

• @ayse3993 → /workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ ls -lh
total 52K
-rw-rw-rw- 1 codespace root 2.9K Oct  7 13:30 CONTRIBUTING.md
-rw-rw-rw- 1 codespace root 11K Oct  7 13:30 README.md
-rw-rw-rw- 1 codespace root 1.4K Oct  7 13:30 TEAM.md
-rw-rw-rw- 1 codespace root 1.5K Oct  7 13:30 TEAM_TEMPLATE.md
drwxrwxrwx+ 2 codespace root 4.0K Oct  7 13:30 config
drwxrwxrwx+ 3 codespace root 4.0K Oct  7 13:30 data
drwxrwxrwx+ 2 codespace root 4.0K Oct  7 13:30 evidence
drwxrwxrwx+ 2 codespace root 4.0K Oct  7 13:30 labs
drwxrwxrwx+ 2 codespace root 4.0K Oct  7 13:30 readiness
-rw-rw-rw- 1 codespace root 133 Oct  7 13:30 requirements.txt
drwxrwxrwx+ 2 codespace root 4.0K Oct  7 13:30 src
• @ayse3993 → /workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ head -n 5 data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
• @ayse3993 → /workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ tail -n 5 data/sample/prices.csv
2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0

```

Optional 3 :

```
@ayse3993 → /workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ find . -type f
./README.md
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md
./evidence/README.md
./requirements.txt
./TEAM_TEMPLATE.md
./src/main.py
./labs/README.md
./labs/TD02_PYTHON_CSV_JSON.md
./labs/TD01_BOOTSTRAP_LINUX.md
./labs/TD03_GIT_LOCAL_WORKFLOW.md
./td1.pdf
./data/sample/bloomberg_reference_sample.json
./data/sample/prices.csv
./data/sample/bloomberg_reference_expected.json
./data/sample/instruments.json
./CONTRIBUTING.md
./config/settings.yml
./.gitignore
./.git/description
./.git/packed-refs
./.git/index
./.git/hooks/pre-push
./.git/hooks/push-to-checkout.sample
./.git/hooks/sendemail-validate.sample
./.git/hooks/commit-msg.sample
./.git/hooks/post-update.sample
./.git/hooks/fsmonitor-watchman.sample
./.git/hooks/pre-rebase.sample
./.git/hooks/pre-merge-commit.sample
```

```
./.git/hooks/pre-rebase.sample
./.git/hooks/pre-merge-commit.sample
./.git/hooks/pre-receive.sample
./.git/hooks/applypatch-msg.sample
./.git/hooks/post-commit
./.git/hooks/update.sample
./.git/hooks/post-merge
./.git/hooks/pre-applypatch.sample
./.git/hooks/prepare-commit-msg.sample
./.git/hooks/pre-push.sample
./.git/hooks/post-checkout
./.git/hooks/pre-commit.sample
./.git/objects/da/2bd23658110a837276802d20d9fd29ca726a60
./.git/objects/45/7c1a8f27a2b674db9e40a71ea19b3c8577107c
./.git/objects/3c/4bff1627e65c62f81a6c7a57ff3b37d7598008
./.git/objects/d7/1a06b1b3ec81192eee7c9d0125e934ec0e4ace
./.git/objects/dc/46bc2e5b1c763effbb282c14d88ba52b132780
./.git/objects/24/bf1511c729955b9ec77031deaca07df2e94339
./.git/objects/14/55b82068b57097edc3478f86cad425181176a7
./.git/objects/pack/pack-3a2956303bdd3bb2c2f21f411a1cf11b2c3b466c.pack
./.git/objects/pack/pack-3a2956303bdd3bb2c2f21f411a1cf11b2c3b466c.idx
./.git/objects/pack/pack-3a2956303bdd3bb2c2f21f411a1cf11b2c3b466c.rev
./.git/objects/64/7d715da9b5865cc95efa3faa4a45e67912b4b4
./.git/objects/d6/6fbd9f27b573bd3402cb1930d468209883d6fa
./.git/objects/83/bf5fb811ba3ef54d40d3f25459a537b1d0109d
./.git/objects/2c/c2a6340cba8daddcfab56df28434a128b8af85
./.git/objects/ba/5c9f4940c5f6dd78f9788f0116e2966297e1dd
./.git/objects/52/07db82cdabe395615768ee6adcaec1c00da6d
./.git/objects/52/77e1b94c32db506a37063100527def9b258dec
./.git/objects/c3/66828d6497c00eba66c17053287fa1bb92fad2
./.git/objects/7e/caf7eb4e4a4802a12d04aebd6cca35237bfff6b
```

```

./.git/objects/c3/66828d6497c00eba66c17053287fa1bb92fad2
./.git/objects/7e/caf7eb4e4a4802a12d04aebd6cca35237bfff6b
./.git/objects/6e/d55788735d61ab1540e56dc395fc0a2bb22281
./.git/objects/4e/1e430479de6feb4f14b78b2f48f7dfde4d5886
./.git/objects/fc/11001927d154f8ed13891ef676a34b39d4f840
./.git/objects/fc/81a97574a2555836de779f8ce2cc54ed559c5f
./.git/objects/7c/43ecf7088dbc363a571a57cf9fb03b4fbaeabe
./.git/objects/e6/9de29bb2d1d6434b8b29ae775ad8c2e48c5391
./.git/objects/cf/babe4596e3db8e3c7b5d801deb509af6c0ebd6
./.git/objects/55/82d2168f4fcd0f39396f6386244af7947e1c50
./.git/objects/86/00efa9dc3b2ebda55611727a6944dbb1160603
./.git/objects/91/ddf907f9ea4ea40f1c2d6db1dcf54623410cdf
./.git/objects/20/b940264b0b1ccc06bf2b2017beefb76fde5d24
./.git/objects/c7/afec5d34aa93738b9b5227bcacf8eb28cec63f0
./.git/objects/bb/80ff648351adb39528767b1d86d42db22f3369
./.git/objects/4c/9dd98b1df63525b95a94221c366fb786a7fb8b
./.git/objects/84/76ae5f6c4534dffba2508ec863d0bfcfeba0a7
./.git/objects/b6/ac4a901167f08ad9d8c6d63062eadd050ed009
./.git/objects/75/ae0acf22f112d779d278d29d8daebd21808a78
./.git/objects/9f/d6c8a18fbc6b648d35f9e70483bc03043d16e8
./.git/objects/85/13e7a00488f4473dddd817bfe80db66c969ff7
./.git/objects/bf/b94a6112ada1f2183bdfbc499cf0769a7d3933
./.git/config
./.git/refs/heads/ayseguneyusu
./.git/refs/remotes/origin/main
./.git/refs/remotes/origin/PAGET_Mael
./.git/refs/remotes/origin/HEAD
./.git/refs/remotes/origin/adjasy
./.git/HEAD
./.git/logs/refs/heads/ayseguneyusu
./.git/logs/refs/remotes/origin/main

```

```

./.git/logs/refs/remotes/origin/main
./.git/logs/refs/remotes/origin/PAGET_Mael
./.git/logs/refs/remotes/origin/HEAD
./.git/logs/refs/remotes/origin/adjasy
./.git/logs/HEAD
./.git/info/exclude
./.git/FETCH_HEAD
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneyusu) $ grep -R "AAPL" .
./README.md:Instrument : AAPL - Apple Inc.
./README.md:AAPL
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:AAPL - Apple Inc.
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:AAPL: 21
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:grep -c AAPL data/sample/prices.csv
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:AAPL
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:[x] AAPL observation count is 21
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:grep AAPL data/sample/prices.csv
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:AAPL
./TEAM_TEMPLATE.md:- Instrument: AAPL - Apple Inc.
./labs/TD02_PYTHON_CSV_JSON.md:Instrument : AAPL - Apple Inc.
./labs/TD02_PYTHON_CSV_JSON.md:[ ] separates AAPL and SP500 rows
./labs/TD02_PYTHON_CSV_JSON.md:AAPL -> 250.00
./labs/TD02_PYTHON_CSV_JSON.md:AAPL -> 266.20
./labs/TD02_PYTHON_CSV_JSON.md:Do not duplicate one display implementation for AAPL and another for SP500.
./labs/TD02_PYTHON_CSV_JSON.md:AAPL - Apple Inc.
./labs/TD02_PYTHON_CSV_JSON.md:AAPL - Apple Inc.
./labs/TD02_PYTHON_CSV_JSON.md:[ ] 21 observations for AAPL
./labs/TD02_PYTHON_CSV_JSON.md:[ ] AAPL has 21 observations
./labs/TD01_BOOTSTRAP_LINUX.md:Instrument : AAPL - Apple Inc.
./labs/TD01_BOOTSTRAP_LINUX.md:grep AAPL data/sample/prices.csv
./labs/TD01_BOOTSTRAP_LINUX.md:AAPL - Apple Inc.
./labs/TD01_BOOTSTRAP_LINUX.md:AAPL: 21

```



```
./labs/TD01_BOOTSTRAP_LINUX.md:AAPL: 21
./labs/TD01_BOOTSTRAP_LINUX.md:instrument : AAPL
./labs/TD01_BOOTSTRAP_LINUX.md:grep -c AAPL data/sample/prices.csv
./labs/TD01_BOOTSTRAP_LINUX.md:grep -R "AAPL" .
./data/sample/bloomberg_reference_sample.json: "canonical_ticker": "AAPL",
./data/sample/bloomberg_reference_sample.json: "provider_identifier": "AAPL US Equity",
./data/sample/prices.csv:2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
./data/sample/prices.csv:2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
./data/sample/prices.csv:2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
./data/sample/prices.csv:2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
./data/sample/prices.csv:2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
./data/sample/prices.csv:2026-09-09,AAPL,254.00,255.70,252.80,254.20,38000000
./data/sample/prices.csv:2026-09-10,AAPL,254.30,256.60,253.10,255.10,39500000
./data/sample/prices.csv:2026-09-11,AAPL,254.20,256.20,253.00,254.70,41000000
./data/sample/prices.csv:2026-09-14,AAPL,256.60,258.30,255.40,256.80,42500000
./data/sample/prices.csv:2026-09-15,AAPL,257.20,259.50,256.00,258.00,44000000
./data/sample/prices.csv:2026-09-16,AAPL,256.70,258.70,255.50,257.20,38000000
./data/sample/prices.csv:2026-09-17,AAPL,259.20,260.90,258.00,259.40,39500000
./data/sample/prices.csv:2026-09-18,AAPL,259.30,261.60,258.10,260.10,41000000
./data/sample/prices.csv:2026-09-21,AAPL,259.10,261.10,257.90,259.60,42500000
./data/sample/prices.csv:2026-09-22,AAPL,261.10,262.80,259.90,261.30,44000000
./data/sample/prices.csv:2026-09-23,AAPL,261.20,263.50,260.00,262.00,38000000
./data/sample/prices.csv:2026-09-24,AAPL,263.00,265.00,261.80,263.50,39500000
./data/sample/prices.csv:2026-09-25,AAPL,262.60,264.30,261.40,262.80,41000000
./data/sample/prices.csv:2026-09-28,AAPL,263.30,265.60,262.10,264.10,42500000
./data/sample/prices.csv:2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
./data/sample/prices.csv:2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
./data/sample/bloomberg_reference_expected.json: "ticker": "AAPL",
./data/sample/bloomberg_reference_expected.json: "ticker": "AAPL",
./data/sample/bloomberg_reference_expected.json: "ticker": "AAPL",
./data/sample/bloomberg_reference_expected.json: "ticker": "AAPL",
```

Optional 4 :

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ git status
On branch ayseguneysu
Your branch is up to date with 'origin/ayseguneysu'.

nothing to commit, working tree clean
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ git log --oneline
0866299 (HEAD -> ayseguneysu, origin/ayseguneysu) Add files via upload
5207db8 first td session
ba5c9f4 Update Readme
9fd6c8a Update Readme
bfb94a6 init projecttemplate
```

Td 2 :

Part 1 :

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ code src/main.py
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
```



```

from pathlib import Path
import csv
import json

DATA_DIR = Path("data/sample")

# These starter values mirror config/settings.yml.
# settings.yml is a human-readable configuration contract in the CORE.
# Parsing YAML is optional and is not required by the 18-hour lab sequence.
LOOKBACK_LABEL = "1 month"
INTERVAL_LABEL = "Daily"

def load_instruments():
    with open(DATA_DIR / "instruments.json", encoding="utf-8") as file:
        return json.load(file)

def load_prices():
    with open(DATA_DIR / "prices.csv", encoding="utf-8") as file:
        return list(csv.DictReader(file))

def filter_prices(prices, ticker):
    return [row for row in prices if row["ticker"] == ticker]

if __name__ == "__main__":
    main()

```

```

@ayse3993 → /workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ cat src/main.py
from pathlib import Path
import csv
import json

DATA_DIR = Path("data/sample")

# These starter values mirror config/settings.yml.
# settings.yml is a human-readable configuration contract in the CORE.
# Parsing YAML is optional and is not required by the 18-hour lab sequence.
LOOKBACK_LABEL = "1 month"
INTERVAL_LABEL = "Daily"

def load_instruments():
    with open(DATA_DIR / "instruments.json", encoding="utf-8") as file:
        return json.load(file)

def load_prices():
    with open(DATA_DIR / "prices.csv", encoding="utf-8") as file:
        return list(csv.DictReader(file))

```

```

def filter_prices(prices, ticker):
    return [row for row in prices if row["ticker"] == ticker]

def main():
    instruments = load_instruments()
    prices = load_prices()

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]

    instrument_prices = filter_prices(prices, instrument["ticker"])
    benchmark_prices = filter_prices(prices, benchmark["ticker"])

    instrument_latest = instrument_prices[-1]
    benchmark_latest = benchmark_prices[-1]

    print("=== MarketPulse ===")
    print()
    print("Instrument")
    print(f"{instrument['ticker']} - {instrument['name']}")
    print(f"Last price: {instrument_latest['close']} {instrument['currency']}")
    print()
    print("Benchmark")
    print(f"{benchmark['ticker']} - {benchmark['name']}")
    print(f"Last level: {benchmark_latest['close']}")

```

```

instrument_prices = filter_prices(prices, instrument["ticker"])
benchmark_prices = filter_prices(prices, benchmark["ticker"])

instrument_latest = instrument_prices[-1]
benchmark_latest = benchmark_prices[-1]

print("=== MarketPulse ===")
print()
print("Instrument")
print(f"{instrument['ticker']} - {instrument['name']}")
print(f"Last price: {instrument_latest['close']} {instrument['currency']}")
print()
print("Benchmark")
print(f"{benchmark['ticker']} - {benchmark['name']}")
print(f"Last level: {benchmark_latest['close']}")
print()
print(f"Period: {LOOKBACK_LABEL}")
print(f"Interval: {INTERVAL_LABEL}")
print()
print("Observations")
print(f"{instrument['ticker']}: {len(instrument_prices)}")
print(f"{benchmark['ticker']}: {len(benchmark_prices)}")

```

```

if __name__ == "__main__":
    main()

```

load_instruments() reads the JSON file, load_prices() reads the CSV file, filter_prices() selects the data for one ticker, and main() runs and coordinates everything. The input files are in the data/sample/ folder.

Part 2 :

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ python src/main.py
<class 'dict'>
dict_keys(['instrument', 'benchmark'])
AAPL
Apple Inc.
USD
SP500
S&P 500
```

```
def main():
    instruments = load_instruments()
    prices = load_prices()

    print(type(instruments))
    print(instruments.keys())

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]

    instrument_prices = filter_prices(prices, instrument["ticker"])
    benchmark_prices = filter_prices(prices, benchmark["ticker"])

    instrument_latest = instrument_prices[-1]
    benchmark_latest = benchmark_prices[-1]

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]

    print(instrument["ticker"])
    print(instrument["name"])
    print(instrument["currency"])

    print(benchmark["ticker"])
    print(benchmark["name"])
```

Part 3 :

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ head data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
```

```
def main():
    instruments = load_instruments()
    prices = load_prices()

    print(type(prices))
    print(type(prices[0]))
    print(prices[0])

    print(prices[0]["close"])
    print(type(prices[0]["close"]))

    close = float(prices[0]["close"])
    print(close)
    print(type(close))

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]
```

```
<class 'list'>
@ayse3993 → /workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ python src/main.py
<class 'list'>
<class 'dict'>
{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low': '248.00', 'close': '250.00', 'volume': '38000000'}
250.00
<class 'str'>
250.0
<class 'float'>
```

Part 4:

```
def main():
    instruments = load_instruments()
    prices = load_prices()

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]

    instrument_prices = filter_prices(
        prices,
        instrument["ticker"],
    )

    benchmark_prices = filter_prices(
        prices,
        benchmark["ticker"],
    )

    print(len(instrument_prices))
    print(len(benchmark_prices))
```

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ python src/main.py
21
21
~
```

Part 5 :

```
● @ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ python src/main.py
250.0
266.2
6600.0
6742.0
```

```
def main():
    instruments = load_instruments()
    prices = load_prices()

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]

    instrument_prices = filter_prices(prices, instrument["ticker"])
    benchmark_prices = filter_prices(prices, benchmark["ticker"])

    print(get_first_close(instrument_prices))
    print(get_last_close(instrument_prices))

    print(get_first_close(benchmark_prices))
    print(get_last_close(benchmark_prices))
```

Part 6 :

```
def display_market_summary(asset, prices, show_currency=True):
    first_close = get_first_close(prices)
    last_close = get_last_close(prices)

    print(f"{asset['ticker']} - {asset['name']}")
    print(f"Observations : {len(prices)}")

    if show_currency:
        print(f"First close : {first_close:.2f} {asset['currency']}")
        print(f>Last close : {last_close:.2f} {asset['currency']}")
    else:
        print(f"First close : {first_close:.2f}")
        print(f>Last close : {last_close:.2f}")
```



```
def main():
    instruments = load_instruments()
    prices = load_prices()

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]

    instrument_prices = filter_prices(prices, instrument["ticker"])
    benchmark_prices = filter_prices(prices, benchmark["ticker"])

    display_market_summary(instrument, instrument_prices)
    print()
    display_market_summary(benchmark, benchmark_prices, False)
```

```
@ayse3993 → /workspaces/esilv-marketpulse-g01-t07 (aysegunneysu) $ python src/main.py
AAPL - Apple Inc.
Observations : 21
First close   : 250.00 USD
Last close    : 266.20 USD

SP500 - S&P 500
Observations : 21
First close   : 6600.00
Last close    : 6742.00
--- MarketPulse ---
```

Part 7/8 :

```
def main():
    instruments = load_instruments()
    prices = load_prices()

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]

    instrument_prices = filter_prices(prices, instrument["ticker"])
    benchmark_prices = filter_prices(prices, benchmark["ticker"])

    print("=== MarketPulse ===")
    print()
    print(f"Period: {LOOKBACK_LABEL}")
    print(f"Interval: {INTERVAL_LABEL}")
    print()

    display_market_summary(instrument, instrument_prices)
    print()
    display_market_summary(benchmark, benchmark_prices, False)

if __name__ == "__main__":
    main()
```

```

@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (aysegunneysu) $ python src/main.py
=== MarketPulse ===

Period: 1 month
Interval: Daily

AAPL - Apple Inc.
Observations : 21
First close  : 250.00 USD
Last close   : 266.20 USD

SP500 - S&P 500
Observations : 21
First close  : 6600.00
Last close   : 6742.00

```

Part 9/10 : It is been done.

```

@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (aysegunneysu) $ python src/main.py
=== MarketPulse ===

Period: 1 month
Interval: Daily

AAPL - Apple Inc.
Observations : 21
First close  : 250.00 USD
Last close   : 266.20 USD

SP500 - S&P 500
Observations : 21
First close  : 6600.00
Last close   : 6742.00

```

Optional 1 :

```

def filter_prices(prices, ticker):
    return [row for row in prices if row["ticker"] == ticker]

def get_first_close(prices):
    return float(prices[0]["close"])

def get_last_close(prices):
    return float(prices[-1]["close"])

def get_first_date(prices):
    return prices[0]["date"]

def get_last_date(prices):
    return prices[-1]["date"]

def display_market_summary(asset, prices, show_currency=True):
    first_close = get_first_close(prices)
    last_close = get_last_close(prices)

```

```
def main():
    instruments = load_instruments()
    prices = load_prices()

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]

    instrument_prices = filter_prices(prices, instrument["ticker"])
    benchmark_prices = filter_prices(prices, benchmark["ticker"])

    print(get_first_date(instrument_prices))
    print(get_last_date(instrument_prices))

    print("=== MarketPulse ===")
    print()
    print(f"Period: {LOOKBACK_LABEL}")
```

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ python src/main.py
2026-09-01
2026-09-30
```

Optional 2 :

```
return prices[-1]["date"]
```

```
def count_positive_volume(prices):
    return sum(int(row["volume"]) > 0 for row in prices)
```

```
def main():
    instruments = load_instruments()
    prices = load_prices()

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]

    instrument_prices = filter_prices(prices, instrument["ticker"])
    benchmark_prices = filter_prices(prices, benchmark["ticker"])

    print(count_positive_volume(instrument_prices))
```

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ python src/main.py
21
=== MarketPulse ===

Period: 1 month
Interval: Daily

AAPL - Apple Inc.
```

Optional 3 :

```
def get_min_close(prices):
    return min(float(row["close"]) for row in prices)
```

```
def get_max_close(prices):
    return max(float(row["close"]) for row in prices)
```

```
def main():
    instruments = load_instruments()
    prices = load_prices()

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]

    instrument_prices = filter_prices(prices, instrument["ticker"])
    benchmark_prices = filter_prices(prices, benchmark["ticker"])

    print(get_min_close(instrument_prices))
    print(get_max_close(instrument_prices))
```

```
Last Close : 0742.00
• @ayse3993 → /workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ python src/main.py
249.8
266.2
```

Optional 4 :

```
def main():
    instruments = load_instruments()
    prices = load_prices()

    print(prices[:3])

    for row in prices[:3]:
        print(row)

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]
```

```
@ayse3993 → /workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ python src/main.py
[{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low': '248.00', 'close': '250.00', 'volume': '38000000'}, {'date': '2026-09-01', 'ticker': 'SP500', 'open': '6592.00', 'high': '6615.00', 'low': '6580.00', 'close': '6600.00', 'volume': '0'}, {'date': '2026-09-02', 'ticker': 'AAPL', 'open': '250.80', 'high': '252.80', 'low': '249.60', 'close': '251.30', 'volume': '39500000'}]
[{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low': '248.00', 'close': '250.00', 'volume': '38000000'}, {'date': '2026-09-01', 'ticker': 'SP500', 'open': '6592.00', 'high': '6615.00', 'low': '6580.00', 'close': '6600.00', 'volume': '0'}, {'date': '2026-09-02', 'ticker': 'AAPL', 'open': '250.80', 'high': '252.80', 'low': '249.60', 'close': '251.30', 'volume': '39500000'}]
```

Optionnal 5 :

```
def main():  
    instruments = load_instruments()  
    prices = load_prices()  
  
    print(f"{266.2:.2f}")  
    print(f"{6742:.2f}")  
  
    instrument = instruments["instrument"]  
    benchmark = instruments["benchmark"]
```

```
● @ayse3993 → /workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ python src/main.py  
266.20  
6742.00
```